Plan the electric car

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use circuit and mechanical evidence to plan a controlled motor-car test.

SUCCESS CRITERIA - I can...

1. Draw the battery-switch-motor circuit.
2. Predict effects of an open switch and a changed connection.
3. Plan one controlled performance comparison.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

The same flux change through the same coil occurs in 1/3 of the time. Compare induced-emf magnitude and name a condition for that comparison.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: P = IV; E = PΔt.
Constant power model. Use approved battery, switch and motor. Disconnect before rewiring; keep wheels clear for first power.

Worked reasoning: A classroom car circuit uses a battery, switch and motor in one closed loop. The switch opens the current path; a chassis drawing must also reserve space for safe wiring and moving parts.

Your turn: Draw the circuit separately from a labeled component-placement plan.



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

03 Read and interpret the model

Draw and label the battery, switch, motor and wires as a complete loop. Mark a location where an open connection stops current.



Circuit symbols: long/short lines = battery; rectangles = resistors; M = motor.

Marzano check - circle: ① ② ③

04 Apply the physics

During a supplied model-car test, the motor draws 0.50 A at 3 V for 6 s and the car travels 5 m. Find electrical input power, input energy and average speed. Complete the table and keep these supplied values separate from your team's later measurements.

Voltage (V)	Current (A)	Time (s)	Distance (m)
3	0.5	6	5

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

Your team changes wheels, battery and axle alignment together and gets a faster run. Can you attribute the change to wheels alone? Explain a better comparison.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Electrons in a wire drift at 0.4 mm/s while a lamp responds quickly after a switch closes. Explain why response time cannot be found by timing one electron across the wire.
Draw the stated circuit and label quantities relevant to the question.

Draw the complete circuit and label shared nodes.

Marzano check - circle: 1 2 3

07 Exit check

Name two reasons electrical input energy will exceed the car’s final translational kinetic energy.

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 6 V motor draws 0.8 A normally and 2.4 A when stalled in supplied test data. Compare input powers and explain a test plan that avoids prolonged stall.

Condition	Voltage (V)	Current (A)	Power (W)
Running	6	0.8	____
Stalled	6	2.4	____

Marzano check - circle: 1 2 3